

Predicting Diabetes from Routine Health Metrics: A Comparative Machine Learning Study

Group 07

Department of Computer Science and Engineering

Course: CSE422 — Artificial Intelligence

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Abstract—Early, low-cost screening for diabetes is valuable because comprehensive diagnostic testing is not always accessible, while routine physiological measurements are inexpensive and widely collected. This report investigates whether such routine measurements – age, body mass index (BMI), temperature, heart rate, blood pressure, cholesterol, gender, blood type, and smoking status – can predict diabetes status, using a dataset of 200,000 individuals. Following exploratory data analysis and pre-processing (missing-value imputation, redundant-feature removal, categorical encoding, and feature scaling), five supervised classifiers – Logistic Regression, Gaussian Naive Bayes, a Decision Tree, K-Nearest Neighbors (KNN), and a Neural Network (Multi-Layer Perceptron) – were trained and evaluated using accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). K-Means clustering was additionally applied without access to the diabetes labels. Exploratory analysis found negligible correlation ($|r| < 0.007$) between every numeric feature and the diabetes label across Pearson, Spearman, and Kendall methods. Consistent with this, all five classifiers performed close to a majority-class baseline (AUC 0.50–0.66), and K-Means clusters showed no meaningful correspondence with true diabetes status (Adjusted Rand Index ≈ 0). KNN achieved the highest AUC (0.656), but this is attributed to sensitivity to local sampling noise rather than genuine signal. These results indicate that, in this dataset, general vital-sign measurements alone carry little predictive information about diabetes status.

Index Terms—Diabetes prediction, machine learning, exploratory data analysis, classification, K-means clustering, neural networks.

I. INTRODUCTION

Diabetes is one of the most common chronic conditions worldwide, and a large share of cases go undiagnosed because comprehensive diagnostic testing (fasting glucose, HbA1c) is not always accessible or affordable. Routine measurements – age, body mass index (BMI), blood pressure, heart rate, cholesterol, body temperature, and smoking status – are far cheaper and are already collected during most general check-ups. If these routine measurements carry meaningful signal about diabetes risk, they could support low-cost, early screening in settings where full diagnostic testing is impractical.

The aim of this project is to use a health-metrics dataset to build and compare several machine learning models – a Neural Network together with several classical classifiers – that predict whether an individual has diabetes from routine metrics. The labels are then set aside entirely and K-Means clustering is applied, to test whether the data separates into

groups that correspond to diabetic and non-diabetic status without supervision.

The target feature is *Diabetes*, which takes the discrete values *Yes* and *No*, making this a binary classification problem. The remainder of this report is organized as follows. Section II describes the dataset. Section III presents the exploratory data analysis. Section IV details pre-processing. Section V describes model training. Section VI reports evaluation results, and Section VII concludes with a discussion of probable causes behind the results and recommendations for future work.

II. DATASET DESCRIPTION

The dataset (`7.csv`) contains 200,000 rows and 13 columns: 12 predictor features and one target variable, *Diabetes*. Table I summarizes the columns. Eight columns are numeric (*Student ID*, *Age*, *Height*, *Weight*, *BMI*, *Temperature*, *Heart Rate*, *Blood Pressure*, *Cholesterol* – nine including the identifier) and four are categorical (*Gender*, *Blood Type*, *Diabetes*, *Smoking*).

TABLE I
DATASET FEATURES

Column	Description
Student ID	Row identifier (not a health measurement)
Age	Age in years
Gender	Male / Female
Height	Height (cm)
Weight	Weight (kg)
Blood Type	A / B / AB / O
BMI	Body Mass Index
Temperature	Body temperature (°F)
Heart Rate	Resting heart rate (bpm)
Blood Pressure	Systolic blood pressure
Cholesterol	Cholesterol level
Diabetes	Target: Yes / No
Smoking	Yes / No

III. EXPLORATORY DATA ANALYSIS

A. Missing Values and Duplicates

Every column contains missing values at a rate of approximately 10%, missing independently across columns; as a result 74.6% of raw rows contain at least one missing value. The raw dataset also contains 7,644 exact duplicate rows.

B. Distribution and Skewness

Age, Height, Weight, Heart Rate, Blood Pressure, and Cholesterol are approximately uniformly distributed over their respective ranges, while Temperature is approximately normally distributed. Skewness for all numeric features falls within ± 0.5 (fairly symmetric), with BMI showing the largest skew at 0.44, consistent with its distribution as a ratio of two variables. An outlier check using the $1.5 \times \text{IQR}$ rule found negligible outliers for most features (0%), with minor exceptions in Temperature (0.72%) and BMI (0.10%).

C. Correlation Analysis

Fig. 1 shows the correlation matrix of numeric features. The only substantial relationship is between BMI and its constituent measurements: $|r| = 0.822$ with Weight and $|r| = 0.551$ with Height, consistent with the definition $\text{BMI} = \text{weight}(\text{kg})/\text{height}(\text{m})^2$. All other pairwise correlations among numeric features are below 0.01 in magnitude.

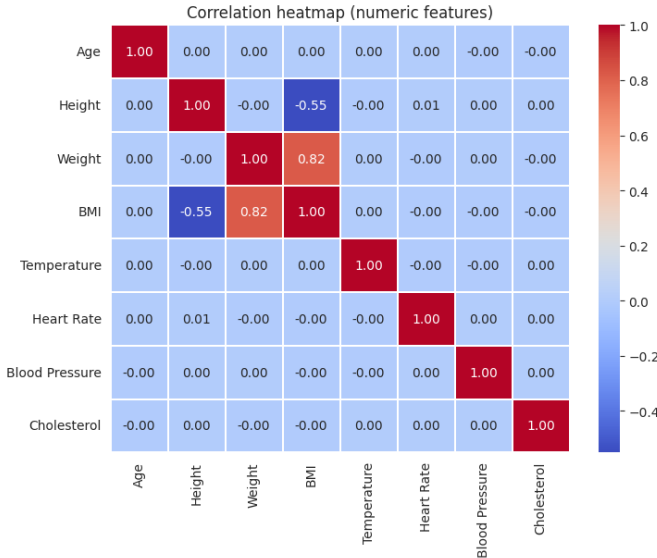


Fig. 1. Correlation heatmap of numeric features. BMI is strongly related to Height and Weight by construction; all other pairs are effectively uncorrelated.

Critically, correlating each numeric feature against the target (Pearson, Spearman, and Kendall methods) yields a maximum $|r|$ of only 0.0067 across every feature and every method – an order of magnitude below what is typically considered even a weak relationship. Group-wise diabetes rates confirm this: the diabetes rate is approximately 10% regardless of smoking status, gender, or blood type.

D. Class Imbalance

As shown in Fig. 2, the target is imbalanced: 89.99% of labelled rows are *No* and 10.01% are *Yes*. This has two consequences addressed later in this report: raw accuracy is a misleading metric (a model that always predicts *No* already scores $\approx 90\%$ accuracy), and class weighting is applied during model training where supported.

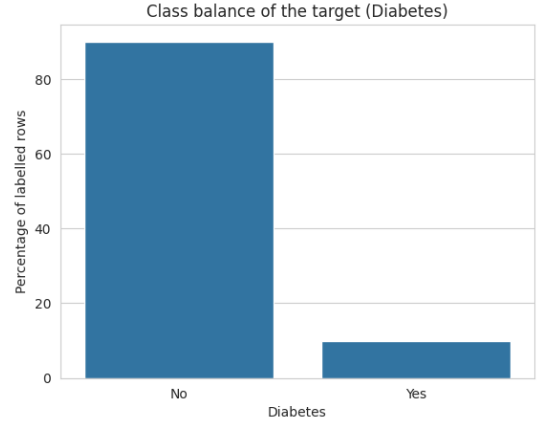


Fig. 2. Class balance of the target variable, Diabetes.

IV. DATA PRE-PROCESSING

Pre-processing was applied in the following order.

Missing target values. Rows with a missing *Diabetes* label (20,000 rows) were dropped, since no ground truth is available to train or evaluate against for these rows.

Duplicates. 7,549 exact duplicate rows (among the remaining 180,000) were removed, leaving 172,451 rows.

Feature selection. *Student ID* was dropped: it has only 96,424 unique values across 172,451 rows and therefore does not even uniquely identify rows, let alone measure health. *Height* and *Weight* were dropped in favor of *BMI*, which is derived directly from them (Section III-C); retaining all three would provide the same information redundantly.

Train/test split. The data was split 80/20 into training and test sets, stratified on *Diabetes* to preserve the $\approx 90/10$ class ratio in both sets (137,960 training rows, 34,491 test rows). All subsequent steps were fit on the training set only and applied to the test set, to avoid data leakage.

Imputation. Remaining missing values were imputed using the median (numeric features) or the most frequent value (categorical features), fit on the training set.

Encoding. *Gender* and *Smoking* were mapped to binary 0/1 values. *Blood Type* (four unordered categories) was one-hot encoded to avoid implying a false ordinal relationship. The target was mapped to 0/1.

Scaling. The six continuous numeric features were standardized (zero mean, unit variance) using training-set statistics, since distance- and gradient-based models (KNN, Logistic Regression, the Neural Network) are sensitive to feature scale.

V. MODEL TRAINING

Five supervised classifiers were trained: Logistic Regression, Gaussian Naive Bayes, a Decision Tree (max depth 8), K-Nearest Neighbors ($k = 15$, distance-weighted), and a Neural Network (Multi-Layer Perceptron, two hidden layers of 64 and 32 units, early stopping). A majority-class *Baseline* classifier was included for reference. `class_weight='balanced'` was applied for Logistic Regression and the Decision Tree to counter the class imbalance identified in Section III-D; Gaussian Naive Bayes and the Neural Network do not support this option in the implementation used.

A. K-Means Clustering

Separately, K-Means was applied to the full (train + test) scaled feature matrix, without access to the *Diabetes* labels. The elbow method (Fig. 3) shows inertia decreasing smoothly with k rather than bending sharply at any one point, suggesting the data lacks strongly separated natural clusters. $k = 2$ was nonetheless selected to allow direct comparison with the two *Diabetes* classes.

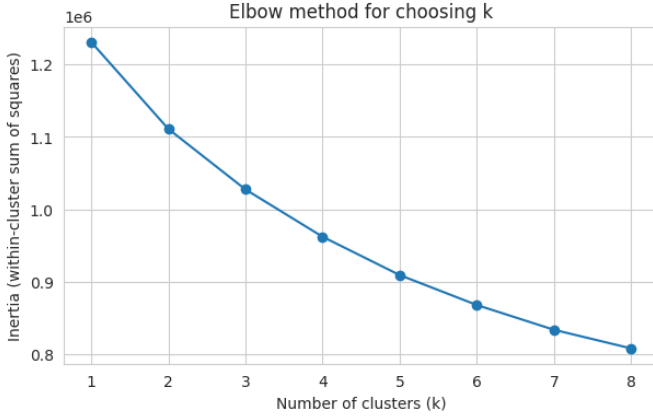


Fig. 3. Elbow method for K-Means. The absence of a sharp elbow suggests weak natural cluster structure.

VI. MODEL EVALUATION

A. Supervised Classifiers

Table II reports accuracy, precision, recall, F1-score, and AUC for all six models on the held-out test set, sorted by AUC. Fig. 4 shows the corresponding ROC curves.

Gaussian Naive Bayes and the Neural Network – the two models without class-weight correction – collapsed to predicting the majority class for every test instance, matching the Baseline exactly on accuracy and AUC. Logistic Regression and the Decision Tree, both class-weighted, achieved higher recall (≈ 0.49 – 0.50) at a heavy cost to precision and overall accuracy (≈ 0.50 – 0.53), the expected trade-off from balancing an imbalanced problem with no strong underlying signal. K-Nearest Neighbors achieved the highest AUC (0.6555) and, notably, the highest precision (0.988) among models that made any positive predictions at all, though its recall remained low (0.119).

B. Cross-Validation

All results above come from a single 80/20 train/test split, which can occasionally give an optimistic or pessimistic read of a model’s true performance. Following the K-Fold Cross-Validation method, each model was re-evaluated using 5-fold *Stratified* K-Fold cross-validation (stratified to preserve the $\approx 90/10$ class ratio in every fold, consistent with Section IV’s stratified split), with a freshly-initialized copy of each model trained and scored on each fold.

Table III reports the mean and standard deviation of AUC across folds, alongside the original single-split AUC for comparison. The cross-validated means agree closely with the

single-split results (differences of 0.005 or less in every case), and the standard deviation across folds is small (0.0036–0.0053) for every model. This confirms the single-split result was not a lucky or unlucky artifact of one particular split: K-Nearest Neighbors consistently outperforms the other models across every fold (Fig. 5), while every other model remains close to the 0.5 baseline throughout.

C. K-Means vs. True Labels

K-Means partitioned the data into clusters of 89,573 and 82,878 points. Comparing these clusters against the true *Diabetes* labels gives an Adjusted Rand Index of -0.00012 and a Normalized Mutual Information of 0.00001 – both indistinguishable from zero, indicating no meaningful correspondence between the discovered clusters and diabetes status. The silhouette score (0.096, computed on a 10,000-row sample) further indicates the two clusters are not well separated to begin with.

VII. CONCLUSION

A. Best Model

By raw accuracy, several models appear strong ($\approx 90\%$), but this only reflects matching the 90% majority class rather than any ability to detect diabetic cases. Ranked by AUC – the metric least distorted by class imbalance – K-Nearest Neighbors is the top-performing model (Table II), though its AUC of 0.656 indicates only modest discriminative ability, and, as discussed below, likely overstates the model’s real-world reliability.

B. Probable Reasons Behind These Results

Five independent pieces of evidence in this study converge on the same explanation. First, the correlation analysis (Section III-C) found no numeric feature with more than $|r| = 0.0067$ correlation with the target, across three different correlation methods. Second, group-wise diabetes rates are statistically indistinguishable across smoking status, gender, and blood type. Third, K-Means, operating with no label information whatsoever, discovered clusters with no relationship to diabetes status ($ARI \approx 0$). Fourth, 5-fold cross-validation (Section VI-B) confirms every model’s performance is stable across different subsets of the data (standard deviation under 0.006 in every case), ruling out the possibility that these are lucky or unlucky single-split results. Fifth, the one classifier with a visibly elevated AUC (KNN) is also the most locally flexible model in the comparison, which makes it prone to fitting incidental noise in dense regions of the training data – a pattern not corroborated by any single feature showing real signal in the EDA, though its consistency across cross-validation folds suggests this is a reproducible property of the dataset rather than a one-off artifact. Taken together, these results indicate that the pre-processing, training, and evaluation pipeline are functioning correctly, but the available features in this dataset carry little intrinsic information about diabetes status. Real-world diabetes risk depends heavily on factors not captured here, most notably blood glucose or HbA1c levels

TABLE II
MODEL COMPARISON ON THE TEST SET (SORTED BY AUC)

Model	Accuracy	Precision	Recall	F1 Score	AUC
K-Nearest Neighbors	0.9115	0.9880	0.1189	0.2122	0.6555
Decision Tree	0.5337	0.1049	0.4851	0.1726	0.5215
Gaussian Naive Bayes	0.8998	0.0000	0.0000	0.0000	0.5078
Logistic Regression	0.5042	0.1015	0.5025	0.1689	0.5072
Neural Network (MLP)	0.8998	0.0000	0.0000	0.0000	0.5050
Baseline (majority class)	0.8998	0.0000	0.0000	0.0000	0.5000

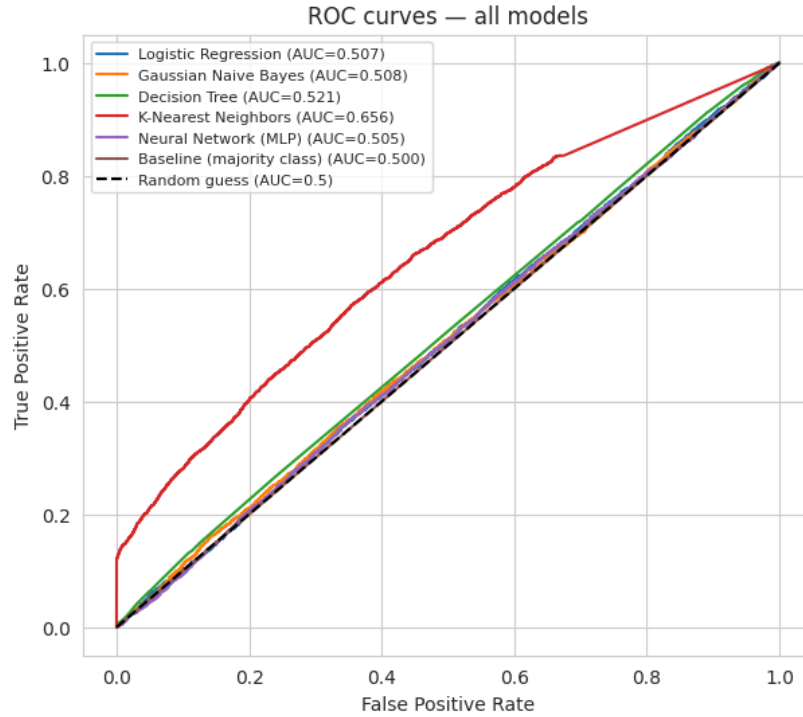


Fig. 4. ROC curves for all supervised models. All curves lie close to the diagonal (AUC = 0.5, random guessing), with K-Nearest Neighbors showing the largest, though still modest, deviation.

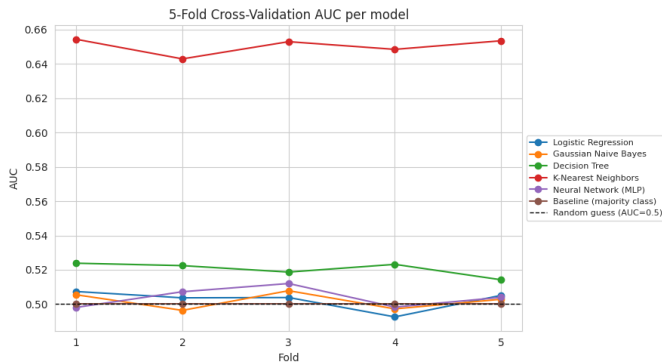


Fig. 5. AUC per model across 5 stratified cross-validation folds. K-Nearest Neighbors remains consistently separated from the other models across every fold.

TABLE III
5-FOLD CROSS-VALIDATION RESULTS (SORTED BY CV MEAN AUC)

Model	CV Mean AUC	CV Std AUC	Single-Split AUC
K-Nearest Neighbors	0.6504	0.0043	0.6555
Decision Tree	0.5204	0.0036	0.5215
Neural Network (MLP)	0.5039	0.0053	0.5050
Logistic Regression	0.5025	0.0051	0.5072
Gaussian Naive Bayes	0.5019	0.0045	0.5078
Baseline (majority class)	0.5000	0.0000	0.5000

C. Recommendations

Future work could: (i) incorporate more directly relevant clinical features, particularly fasting glucose or HbA1c; (ii) collect a larger number of positive (diabetic) cases, since only 10% of the current sample is positive; (iii) apply resampling techniques such as SMOTE in addition to class weighting; and (iv) avoid selecting a "best" model from a single metric in isolation, cross-checking any leaderboard result against exploratory findings, as done in this report.

and family history, so a model built only from general vital signs has a low ceiling on achievable performance regardless of the algorithm used.

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